STANDARD OPENS OF AFFINE SCHEMES ARE AFFINE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1189-1199, following the statement at 1182-1187.

Allowed standard prerequisites (editorial).

- Standard opens as homeomorphic images of localizations. Algebra, Lemma algebra-lemma-standard-open (cross-chapter, cited at ~1193). Spec(R_f) -> Spec(R) is a homeomorphism onto D(f).
- Open immersion image identification. lemma-open-immersion (card L26384, cited at ~1198). Identifies D(f) with U as open subspace.

Source reference. Original reference 00E4: focused-seeds/geometry/S9-schemes-standard-open-affine/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01HH: sources/stacks-schemes/pinned-original-source.tex, source lines 458–458. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let Y be an affine scheme. Let $f \in \Gamma(Y, \mathcal{O}_Y)$. The open subspace $D(f)$ is an affine scheme.*

Proof. We may assume that $Y = \mathrm{Spec}(R)$ and $f \in R$. Consider the morphism of affine schemes $\phi : U = \mathrm{Spec}(R_f) \to \mathrm{Spec}(R) = Y$ induced by the ring map $R \to R_f$. By Algebra, Lemma 00E4 we know that it is a homeomorphism onto $D(f)$. On the other hand, the map $\phi^{-1}\mathcal{O}_Y \to \mathcal{O}_U$ is an isomorphism on stalks, hence an isomorphism. Thus we see that ϕ is an open immersion. We conclude that $D(f)$ is isomorphic to U by Lemma 01HH. □

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.